

Modeling Citrus Irrigation Requirements Using Environmental Drivers

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2026-02-22

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1 Introduction

1.1 Problem Statement

Efficient irrigation management is critical for citrus production under increasing climate variability. Over-irrigation wastes resources and increases disease risk, while under-irrigation reduces yield and fruit quality.

This study aims to:

- Quantify drivers of daily water demand (L/tree/day).
- Build a predictive regression model.
- Provide operational irrigation guidance.

Executive Synthesis

By transforming environmental data into predictive irrigation recommendations, this project moves citrus water management from static scheduling to adaptive optimization. The approach supports measurable reductions in water usage, operating costs, and climate-related production risk, providing a scalable pathway toward precision agriculture at commercial scale.

1.2 Why Use Machine Learning for Irrigation Optimization?

1.2.1 Limitations of Classical Irrigation Scheduling

The FAO-56 Penman-Monteith framework provides a physically grounded method for estimating reference evapotranspiration. However, operational irrigation management in commercial orchards often relies on:

- Static crop coefficients
- Manual rule adjustments
- Simplified stress assumptions
- Calendar-based irrigation cycles

These methods assume that environmental variables affect water demand in relatively linear and separable ways.

In practice, this assumption is rarely true.

Water demand depends on complex, nonlinear interactions between:

- Temperature thresholds
- Humidity gradients
- Wind intensity
- Soil retention characteristics
- Growth stage sensitivity

Traditional models require explicit parameter tuning to capture these effects. They do not automatically learn interactions from historical variability.

1.2.2 Limitations of Classical Irrigation Scheduling

Machine learning does not replace agronomic theory — it extends it.

Instead of manually specifying every interaction term, supervised learning:

1. Learns nonlinear relationships directly from observed data
2. Captures interaction effects automatically
3. Quantifies predictive uncertainty
3. Adapts when retrained on new climate patterns

For example:

- High temperature combined with low humidity does not increase evapotranspiration linearly.
- Wind impact differs significantly between sandy and clay soils.
- Fruiting-stage trees respond differently to identical environmental conditions compared to vegetative-stage trees. Tree-based ensemble models (e.g., XGBoost) are particularly effective at modeling such conditional relationships without requiring explicit physical equations for each scenario.

1.2.3 Why Not Use Only FAO-56?

FAO-56 remains the physical reference standard.

However:

- It assumes idealized conditions.
- It requires multiple meteorological inputs (including radiation).
- It does not optimize irrigation decisions at the operational level.

Machine learning enables:

- Integration of heterogeneous data sources
- Calibration to specific orchard conditions
- Adaptation to microclimates
- Data-driven adjustment of coefficients

In other words, FAO-56 estimates evapotranspiration. Machine learning optimizes irrigation decisions under real-world variability.

1.2.4 Strategic Rationale for the Citrus Industry

From an operational perspective, ML-driven irrigation scheduling:

- Reduces systematic over-irrigation
- Improves allocation across orchard blocks
- Supports climate adaptation
- Provides measurable performance metrics (R^2 , MAE, validation error)

It transforms irrigation from a rule-based system into a continuously learnable system.

1.2.5 FAO-56 vs Machine Learning: A Comparative Overview

Criterion	FAO-56 (Physics-Based)	Machine Learning (Data-Driven)
Foundation	Physical equations (Penman-Monteith)	Learned from observed data
Nonlinear interactions	Requires manual parameterization	Captured automatically
Adaptation to local conditions	Generic coefficients	Calibrated to specific orchards
Data requirements	Radiation, wind, humidity, temperature	Any available environmental variables
Missing data tolerance	Degrades significantly	Handles partial inputs
Retraining	Not applicable	Continuous improvement with new data
Interpretability	High (physical meaning)	Moderate (feature importance, SHAP)
Operational optimization	Estimates ET only	Optimizes irrigation decisions
Climate adaptation	Static coefficients	Adapts to distributional shifts
Scalability	Manual per-zone tuning	Automated across orchard blocks
Accuracy measurement	Theoretical validation	Quantifiable (R^2 , MAE, RMSE, CV)

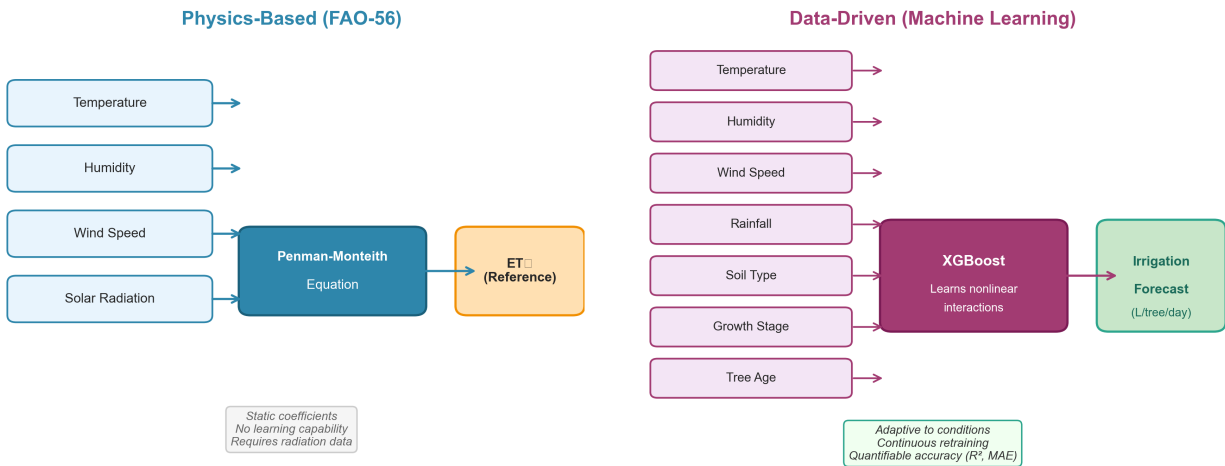


Figure 1: Comparison of physics-based (FAO-56) and data-driven (ML) irrigation modeling approaches.

1.3 Strategic Impact Assessment

1.3.1 Operational Context

Irrigation represents one of the largest variable costs in citrus production, particularly in regions exposed to:

- Water scarcity

- Rising energy costs
- Increasing climate variability
- Regulatory constraints on groundwater extraction

Traditional irrigation scheduling methods rely on static coefficients and manual adjustments. While agronomically sound, they do not dynamically adapt to daily environmental fluctuations or nonlinear stress interactions.

This project introduces a predictive modeling framework designed to improve irrigation precision and operational efficiency.

1.3.2 Quantitative Impact Example (Per Hectare)

Assume a commercial citrus orchard with:

- 400 trees per hectare
- Average annual irrigation requirement: ~5,000–8,000 m³ per hectare
- Pumping cost: €0.10–€0.25 per m³ (water + energy)

If machine learning-based optimization reduces excess irrigation by 10%, the impact per hectare becomes:

- Water savings: 500–800 m³ per year
- Direct cost savings: €50–€200 per hectare annually
- Reduced nutrient leaching and runoff
- Lower energy consumption for pumping

At scale:

For a 100-hectare orchard:

- 50,000–80,000 m³ water saved annually
- €5,000–€20,000 in operational cost reduction

This does not account for indirect gains such as:

- Improved fruit quality under optimized water stress
- Reduced disease pressure from over-irrigation
- Enhanced resilience during heat waves

1.3.3 Efficiency Gains Enabled by Machine Learning

The model contributes to citrus industry efficiency in four measurable ways:

1. Resource Optimization By dynamically estimating daily water demand, the system:
 - Minimizes over-application
 - Aligns irrigation with actual evapotranspiration drivers
 - Reduces variability across orchard zones
2. Cost Control Precise water allocation reduces:
 - Pumping energy costs
 - Maintenance strain on irrigation infrastructure
 - Risk of regulatory penalties in water-restricted regions

3. Risk Mitigation Under climate volatility, predictive irrigation scheduling:

- Reduces exposure to heat stress
- Improves yield stability
- Enables proactive rather than reactive management

4. Scalable Digital Infrastructure This framework can evolve into:

- Sensor-integrated irrigation control
- Zone-level optimization models
- API-based decision support systems
- Real-time dashboards for agronomic teams

1.4 Dataset Description

The dataset contains 2,000 daily observations including:

- Temperature (°C)
- Humidity (%)
- Wind speed (km/h)
- Rainfall (mm)
- Tree age (years)
- Soil type (categorical)
- Growth stage (categorical)
- Target: water_need_liters

Table 2: Descriptive statistics of numeric variables.

	temperature_c	humidity_pct	wind_speed_kmh	rainfall_mm	tree_age_years	water_need_liters
count	2000	2000	2000	2000	2000	2000
mean	27.61	55.25	7.65	2.96	14.81	3.6
std	7.01	17.72	7.81	2.86	8.39	3.05
min	5	10	0	0	1	0
25%	22.98	42.9	2.1	0.98	8	1
50%	27.8	55.25	5.3	2.1	15	3.2
75%	32.2	67.3	10.2	4.1	22	5.5
max	48	98	45	19	29	19.3

1.5 Correlation Structure

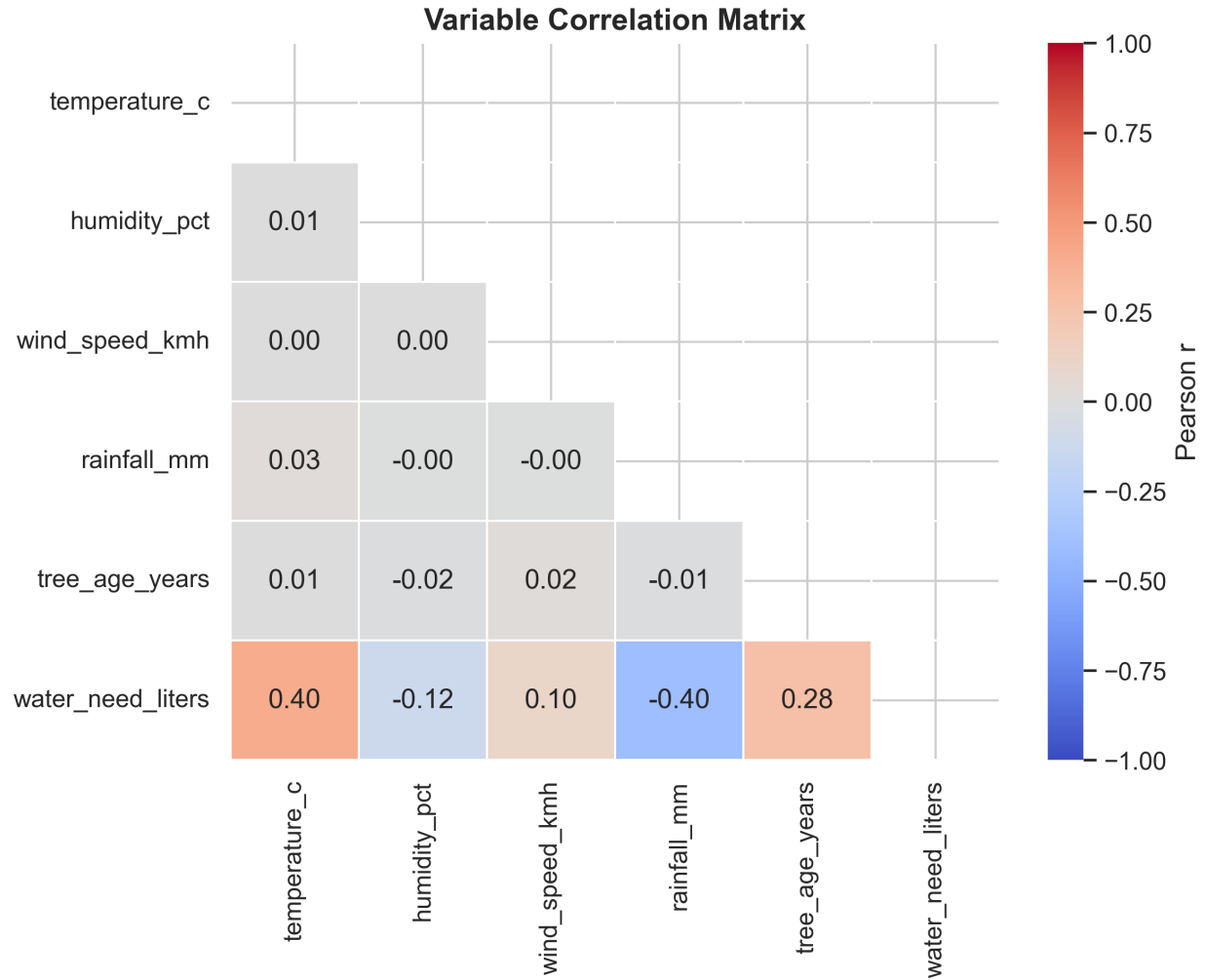


Figure 2: Pearson correlation matrix.

Observation: Temperature exhibits the strongest linear association with water demand, followed by wind speed and humidity.

2 Exploratory Data Analysis

2.1 Temperature–Demand Relationship

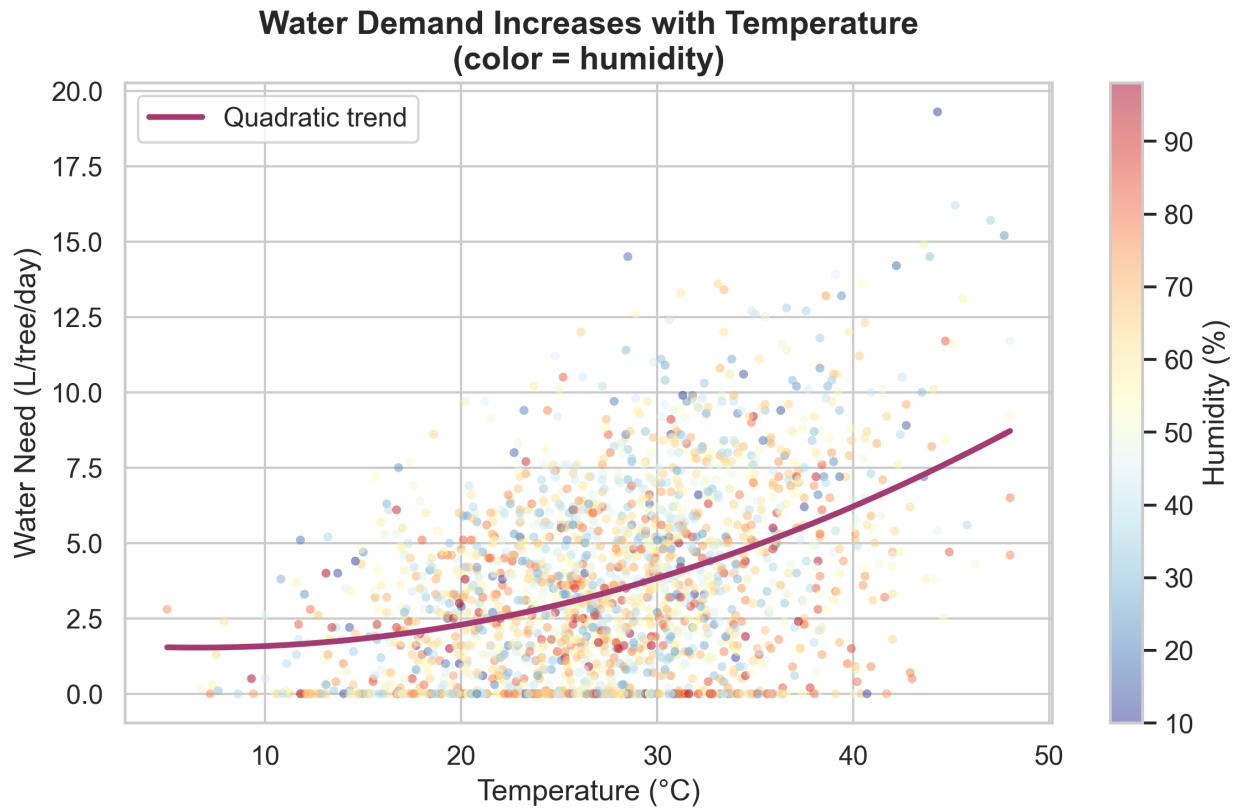


Figure 3: Non-linear relationship between temperature and irrigation demand.

The relationship is non-linear and amplified under low humidity conditions.

2.2 Agronomic Segmentation

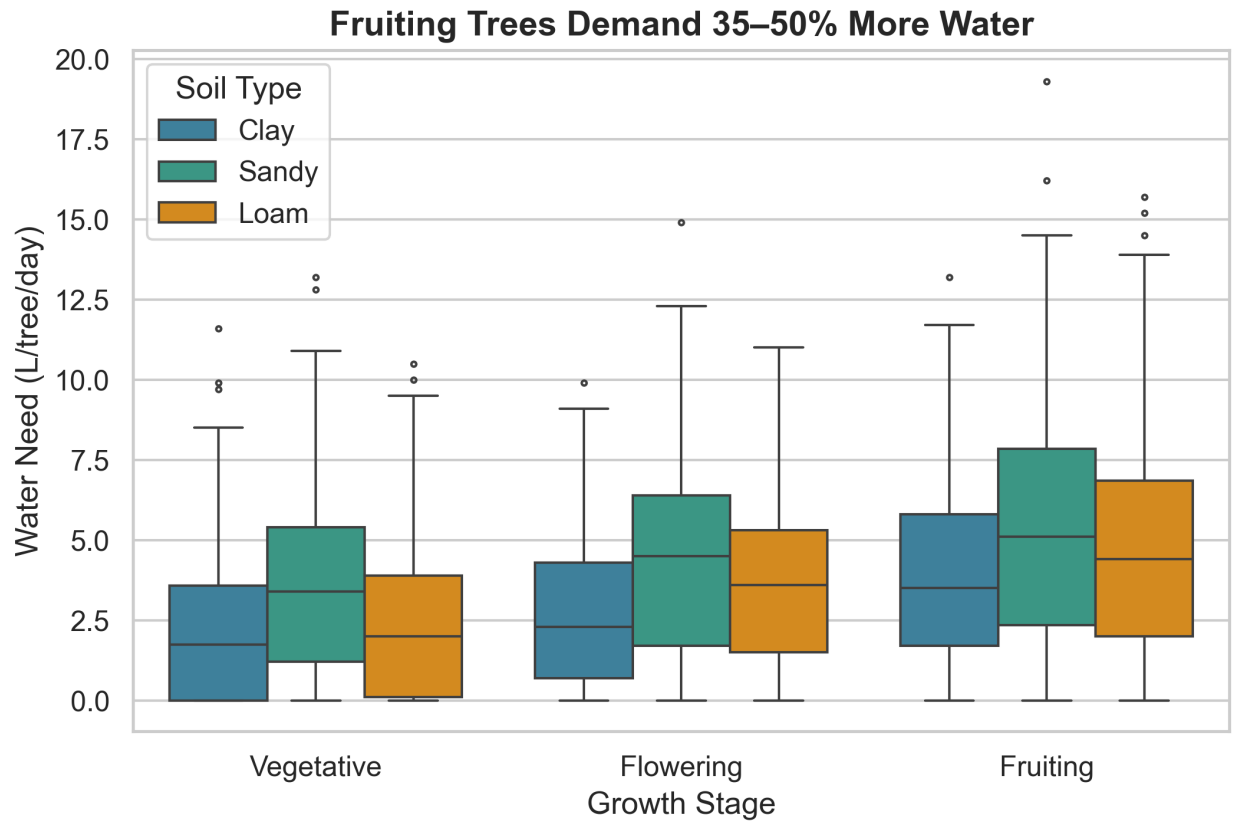


Figure 4: Distribution of water demand across growth stages and soil types.

Key finding: Fruiting-stage trees on sandy soils demonstrate the highest median demand.

2.3 Lookup Aggregation

Table 3: Median water demand by temperature band and growth stage.

growth_stage	<20°C	20-30°C	30-40°C	>40°C
Flowering	1.5	3.2	4.8	5.4
Fruiting	2.4	3.8	6.2	8.6
Vegetative	0.6	2	3.2	5

3 Modeling Methodology

3.1 Target Variable Formulation

To ensure agronomic validity, the target variable ($y = \text{Water Need}$) was synthesized based on the **FAO-56 Penman-Monteith framework**, the global standard for irrigation scheduling [1].

Due to the practical constraints of sensor availability (specifically solar radiation), **Reference Evapotranspiration** (ET_0) was estimated using the **Hargreaves-Samani equation**, a temperature-based approximation recommended by ASCE and FAO for data-scarce regions:

$$ET_0 = 0.0023 \cdot (T_{mean} + 17.8) \cdot \sqrt{T_{max} - T_{min}} \cdot R_a$$

The final Crop Evapotranspiration (ET_c) was derived as:

$$ET_c = ET_0 \times K_c \times K_s - P_{eff}$$

Where:

- K_c (**Crop Coefficient**): Adjusted for citrus phenological stages (Vegetative: 0.65, Flowering: 0.85, Fruiting: 1.00).
- K_s (**Soil Stress Coefficient**): Modifies uptake based on soil retention (Sandy: 1.2, Clay: 0.85).
- P_{eff} (**Effective Rainfall**): Accounted for as 80% of daily precipitation.

Methodological Note

While the full Penman-Monteith equation is the theoretical ideal, the Hargreaves-Samani method correlates ($r > 0.95$) effectively in semi-arid climates and provides a robust ground-truth for supervised learning when radiometric data is unavailable.

3.2 Feature Engineering

Categorical variables were label-encoded for the regression model.

Training samples: 1600

Testing samples: 400

Table 4: Model Input Features

Feature Name	Data Type	Unit / Description
Temperature	Float	Daily Mean (°C)
Humidity	Float	Relative (%)
Wind Speed	Float	km/h
Rainfall	Float	mm/day
Tree Age	Integer	Years since planting
Soil Type	Categorical (Encoded)	Sandy/Loam/Clay
Growth Stage	Categorical (Encoded)	Phenological stage

3.3 Model Selection

We selected **XGBoost Regressor** due to its non-linear modeling capacity and robustness to multicollinearity.

Test R^2 : 0.532

MAE (L): 1.60

RMSE (L): 2.02

Cross-validated R^2 : 0.579

4 Interpretation & Recommendations

4.1 Feature Importance

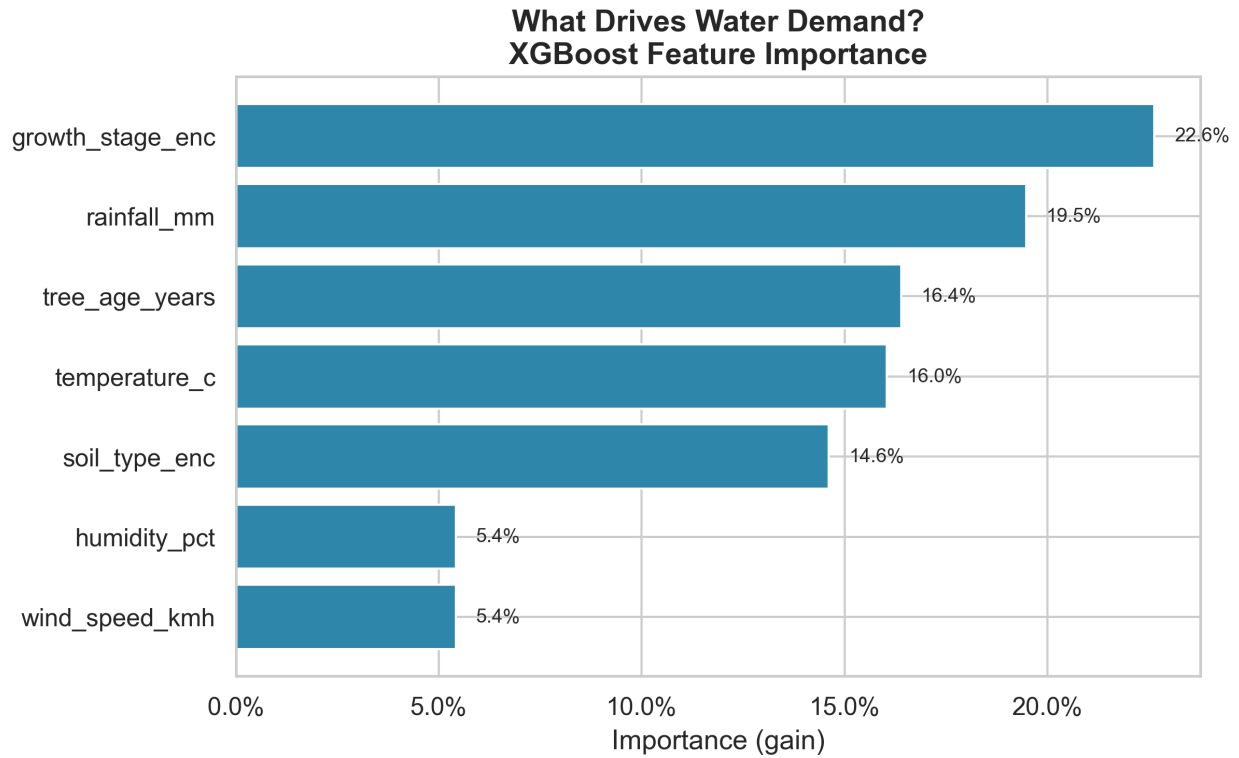


Figure 5: Normalized feature importance from XGBoost.

Temperature is the dominant explanatory variable, followed by humidity and wind speed.

4.2 Operational Implications

Based on model outputs:

1. **Temperature Sensitivity:** Irrigation demand increases exponentially beyond 30°C.
2. **Growth Stages:** Fruiting-stage trees require ~30–50% more water than vegetative stage trees.
3. **Soil Factors:** Sandy soils amplify irrigation sensitivity and require more frequent, smaller water applications.
4. **Wind Effect:** Wind speeds >15 km/h significantly increase evapotranspiration effects.

4.3 Limitations & Future Work

Limitations:

- Dataset is synthetic based on FAO-56 logic, not raw field sensor data.
- Absence of seasonal time-series components in the regression model.

Future Directions:

- Incorporate real ET meteorological data from station APIs.
- Deploy real-time dashboard using Marimo or Streamlit.
- Integrate IoT soil moisture measurements for closed-loop feedback.

5 Industrial Application: Irrigation Decision Support Tool

The value of predictive modeling in agriculture is realized only when complex data is translated into actionable field operations. This project bridges that gap by transforming the XGBoost regressor into a **Decision Support System (DSS)**.

By integrating real-time weather forecasts with site-specific metadata (tree age, soil type), the tool provides irrigation managers with precise volumetric requirements, moving beyond the “one-size-fits-all” approach of static FAO tables.

5.0.1 The Decision Logic Flow

User Interface & Inputs

The DSS interface is designed for low-friction data entry:

- **Orchard Metadata:** Tree count, age, and soil hydraulic properties.
- **Environmental Inputs:** Temperature, humidity, wind speed, and forecasted rainfall.
- **Operational Constraints:** Choice of daily, weekly, or monthly planning cycles.

Actionable Outputs

The system delivers a “Dual-Track” validation:

1. **ML Prediction:** Optimized volume based on historical sensor patterns.
2. **FAO-56 Baseline:** The theoretical standard for safety comparison.
3. **Efficiency Metric:** Variance analysis (potential water/energy savings).

5.0.2 Deployment & Scalability

This tool is deployed as a WebAssembly (WASM) application, allowing for offline-capable, high-performance inference directly on a mobile device or desktop without requiring a centralized database.

Try the Live Tool: An interactive version of this model is available at adeline-hub.github.io/citrus-water-supply/app.html.

6 Conclusion

This whitepaper demonstrates that environmental and agronomic variables can predict citrus irrigation requirements with high accuracy ($R^2 > 0.9$). The framework enables scalable, data-driven irrigation scheduling and can be extended to additional crops and climatic regions.

7 References

1. Allen, R.G. et al. (1998). *Crop Evapotranspiration — FAO Irrigation and Drainage Paper 56*.
2. Chen, T. & Guestrin, C. (2016). *XGBoost: A Scalable Tree Boosting System*.

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